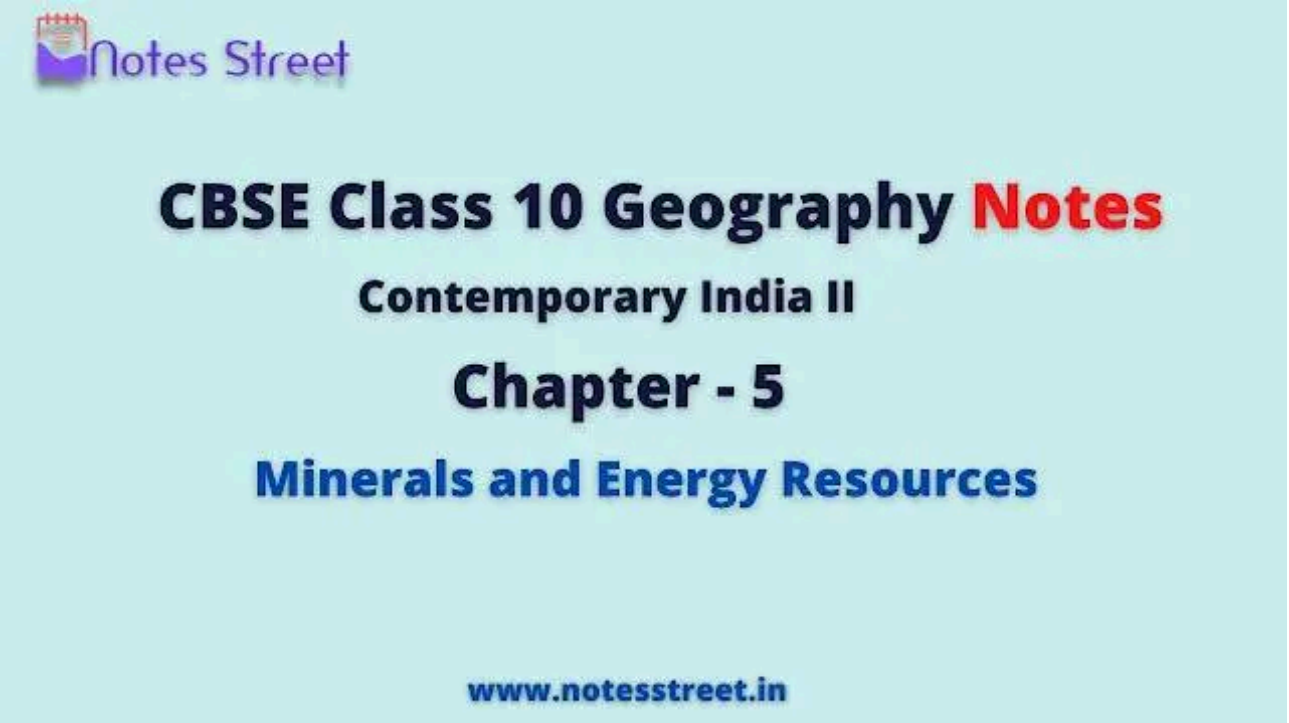

CBSE Class 10 Geography Notes Chapter - 5 Minerals and Energy Resources

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- **Mineral**

(i) Geologists define mineral as a “homogenous, naturally occurring substance with a definable internal structure.”

(ii)Minerals are found in varied forms in nature, ranging from the hardest diamond to the softest talc.

(iii) Rocks are combinations of homogenous substances called minerals.

(iv) Some rocks, for instance limestone, consist of a single mineral only, but majority of the rock consist of several minerals in varying

proportions.

(v) A particular mineral that will be formed from a certain combination of elements depends upon the physical and chemical conditions under which the material forms.

- **Mode of Occurrence of Minerals**

(i) Minerals are usually found in “ores”. The term ore is used to describe any mineral mixed with other elements.

(ii)Minerals generally occur in these forms:

(a) in igneous and metamorphic rocks minerals may occur in the cracks, crevices, faults or joints. The smaller occurrences are called veins and the larger are called lodes. In most cases, they are formed when minerals in liquid/ molten and gaseous forms are forced upward through cavities towards the earth’s surface. They cool and solidify as they rise.

(b) In sedimentary rocks a number of minerals occur in beds or layers. They have been formed as a result of deposition, accumulation and concentration in horizontal strata

(c) Another mode of formation involves the decomposition of surface rocks, and the removal of soluble constituents, leaving a **residual mass of weathered material** containing ores. Bauxite is formed this way.

(d) Certain minerals may occur as alluvial deposits in sands of valley floors and the base of hills. These deposits are called ‘placer deposits’ and generally contain minerals, which are not corroded by water.

(e) The ocean waters contain vast quantities of minerals, but most of these are too widely diffused to be of economic significance.

(iii) It gave the government enormous powers to repress political activities, and allowed detention of political prisoners without trial for two years.

- **Ferrous Minerals**

Ferrous minerals account for about three fourths of the total value of the production of metallic minerals. They provide a strong base for the development of metallurgical industries.

(i) Iron ore: Iron ore is the basic mineral and the backbone of industrial development endowed with fairly abundant resources of iron ore. India is rich in good quality iron ores. Magnetite is the finest iron ore.

The major iron ore belts in India are:

(a) Odisha-Jharkhand belt

(b) Durg-Bastar-Chandrapur belt

(c) Ballari-Chitradurga-Chikkamagaluru-Tumakuru belt

(d) Maharashtra-Goa belt

(ii) Manganese: Manganese is mainly used in the manufacturing of steel and ferro-manganese alloy.

- **Non-Ferrous Minerals**

India's reserves and production of non-ferrous minerals is not very satisfactory. However these minerals, which include copper, bauxite, lead, zinc and gold play a vital role in a number of metallurgical, engineering and electrical industries.

(i) Copper: India is critically deficient in the reserve and production of copper. Being malleable. ductile and a good conductor, copper is mainly used in electrical cables, electronics and chemical industries.

(ii) Bauxite: Though, several ores contain aluminium, it is from bauxite, a clay-like substance that alumina and later aluminium is obtained. Bauxite deposits are formed by the decomposition of a wide variety of rocks rich in aluminium silicates.

- **Non-Metallic Minerals**

Mica is a mineral made up of a series of plates or leaves. It splits easily into thin sheets. These sheets can be so thin that a thousand can be layered into a mica sheet of a few centimeters high Mica can be clear, black, green, red, yellow or brown.

- **Rock Minerals**

Limestone is found in association with rocks composed of calcium carbonates or calcium and magnesium carbonates. It is found in sedimentary rocks of most geological formations. Limestone is the basic raw material for the cement industry and essential for smelting iron ore in the blast furnace.

- **CONSERVATION OF MINERALS**

(i)The geological processes of mineral formation are so slow that the rates of replenishment are infinitely small in comparison to the present rates of consumption.

(ii) Mineral resources are finite and non-renewable. Rich mineral deposits are our country's extremely valuable but short-lived possessions,

(iii) Continued extraction of ores leads to increasing costs as mineral extraction comes from greater depths along with decrease in quality.

(iv) Improved technologies need to be constantly evolved to allow use of low grade ores at costs. Recycling of metals, using scrap metals and other substitutes are steps in consen our mineral resources for the future.

- **Energy Resources**

(i) Energy can be generated from fuel minerals like coal, petroleum, natural gas, uranium and from electricity. Energy resources can be classified as conventional and non-conventional sources

(ii) Conventional sources include: firewood, cattle dung cake, coal, petroleum, naturals gas and electricity (both hydel and thermal).

(iii)Non-conventional sources include solar, wind, tidal, geothermal, biogas and atomic energy Firewood and cattle dung cake are most common in rural India.

(iv)Using dung cake too is being discouraged because it consumes most valuable manure which could be used in agriculture,

- **Conventional Sources of Energy Coal**

(i) In India, coal is the most abundantly available fossil fuel. It provides a substantial part of the nation's energy needs. It is used for power generation, to supply energy to industry as well as for domestic needs.

(ii) Lignite is a low grade brown coal, which is soft with high moisture content

(iii) Coal that has been buried deep and subjected to increased temperatures is bituminous coal

(iv) Metallurgical coal is high grade bituminous coal which has a special value for smelting iron in blast furnaces. Anthracite is the highest quality hard coal.

- **Petroleum**

(i) petroleum or mineral oil is the next major energy source in India after coal It provides fuel for heat and lighting, lubricants for machinery and raw materials for a number of manufacturing industries.

(ii) Petroleum refineries act as a "nodal industry" for synthetic textile, fertiliser and numerous chemical industries.

(iii)Petroleum is also found in fault traps between porous and non-porous rocks. Gas, being lighter usually occurs above the oil.

- **Natural Gas**

(i) Natural gas is an important clean energy resource found in association with or without petroleum. It is used as a source of energy as well as an

industrial raw material in the petrochemical industry.

(ii) Natural gas is considered an environment friendly fuel because of low carbon dioxide emissions and is, therefore, the fuel for the present century.

- **Electricity**

(i) Electricity has such a wide range of applications in today's world that, its per capita consumption is considered as an index of development.

(ii) Electricity is generated mainly in two ways: by running water which drives hydro turbines to generate hydro electricity, and by burning other fuels such as coal, petroleum and natural gas to drive turbines to produce thermal power.

(iii) Hydro electricity is generated by fast flowing water, which is a renewable resource India has a number of multi-purpose projects like the Bhakra Nangal, Damodar Valley corporation, the Kopili Hydel Project etc, producing hydroelectric power.

(iv) Thermal electricity is generated by using coal, petroleum and natural gas. The thermal power stations use non-renewable fossil fuels for generating electricity

- **Non-Conventional Sources of Energy**

There is a pressing need to use renewable energy sources like solar energy, wind, tide, biomass and energy from waste material. These are called non-conventional energy sources.

(i) Nuclear or Atomic Energy: It is obtained by altering the structure of atoms. When such an alteration is made. much energy is released in the form of heat and this is used to generate electric power.

(ii) Solar Energy: Photovoltaic technology converts sunlight directly into electricity. Solar energy is fast becoming popular in rural and remote areas. Some big solar power plants are being established in different parts of India which will minimise the dependence of rural hoseholds on

firewood and dung cakes, which in turn will contribute to environmental conservation and adequate supply of manure in agriculture,

(iii) Wind Power: India has areat potential of wind power. The largest wind farm cluster is located in Tamil Nadu from Nagarcoil to Madurai.

(iv) Biogas:

(a) Shrubs, farm waste, animal and human waste are used to produce biogas for domestic consumption in rural areas.

(b) Decomposition of organic matter yields gas, which has higher thermal efficiency in comparison to kerosene, dung cake and charcoal.

(c) Biogas plants are set up at municipal, cooperative and individual levels. The plants using cattle dung are known as 'Gobar gas plants' in rural India. These provide twin benefit to the farmer in the form of energy and improved quality of manure.

(v) Tidal Energy: Oceanic tides can be used to generate electricity. Floodgate dams are built across inlets. During high tide water flows into the inlet and gets trapped when the gate is closed. After the tide falls outside the flood gate, the water retained by the floodgate flowe back to the sea via a pipe that carries it through a power-generating turbine.

(vi) Geo Thermal Energy: Geo thermal energy refers to the heat and electricity produced by using the heat from the interior of the Earth. Geothermal energy exists because, the Earth grows progressively hotter with increasing depth.

- **Conservation of Energy Resources**

(i) Energy is a basic requirement for economic development. Every sector of the national economy – agriculture, industry, transport, commercial and domestic- needs inputs of energy.

(ii) The economic development plans implemented since Independence necessarily required increasing amounts of energy to remain operational

(iii) As a result, consumption of energy in all forms has been steadily rising all over the country

IMPORTANT TERMS

- Minerals: A solid, naturally occurring inorganic substance.
- Homogenous similar in position, structure, and evolutionary origin but not necessarily in function.
- Igneous Rocks or magmatic rock, is one of the three main rock types. The others being sedimentary and metamorphic, Igneous rock is formed through the cooling and solidification of magma or lava.
- Metamorphic Rocks: Rock that was once one form of rock but has changed to another under the influence of heat, pressure, or some other agent without passing through a liquid phase.
- Peninsular Rocks also known as South Indian rock is a common species of magma found on rocky hills in south India. An allied species of peninsular rock is found in the Eastern Ghats
- Natural Gas: Flammable gas, consisting largely of methane and other hydrocarbons, occurring naturally underground (often in association with petroleum) and used as fuel.
- Geologists: Geology is an earth science. An expert or a student of geology is known as a geologist. Hydro Electricity is the term referring to electricity generated by hydropower, the production of electric power through the use of the gravitational force of falling or flowing water.
- Thermal Power is power generated by using heat.
- Solar Energy: Energy obtained from the sun's radiations.
- Wind Power: power obtained by harnessing the energy of the wind.
- Non-metallic Minerals: Non-metallic Mineral reserves consist of stone quarries and clay and sand pits; chemical and fertiliser mineral deposits, salt deposits, deposits of quartz, gypsum, natural gem stones, asphalt and

bitumen, peat and other non-metallic minerals other than coal and petroleum.

- Alluvial Deposits: Sand, silt, clay, gravel, or other matter deposited by flowing water, as in a riverbed, floodplain, delta. These materials deposited by rivers are called alluvial deposits.
- Ferrous Minerals: Minerals that contain iron. These minerals play a very important role in the development of the metallurgical industries of our country and contain iron. They are iron based.
- Metallurgical Industries: The industry, associated with extracting metals from ores, refining them for use, and creating alloys and useful objects from them.
- CNG Compressed natural gas is a fuel which can be used in place of gasoline (petrol), Diesel fuel and propane/LPG, CNG combustion produces fewer undesirable gases than the fuels mentioned above
- Bio Gas: Gaseous fuel, especially methane, produced by the fermentation of organic matter.
- Tidal Energy is a form of hydropower that converts the energy of the tides into electricity or other useful forms of power. The tide is created by the gravitational effect of the sun and the moon on the earth causing cyclical movement of the seas.

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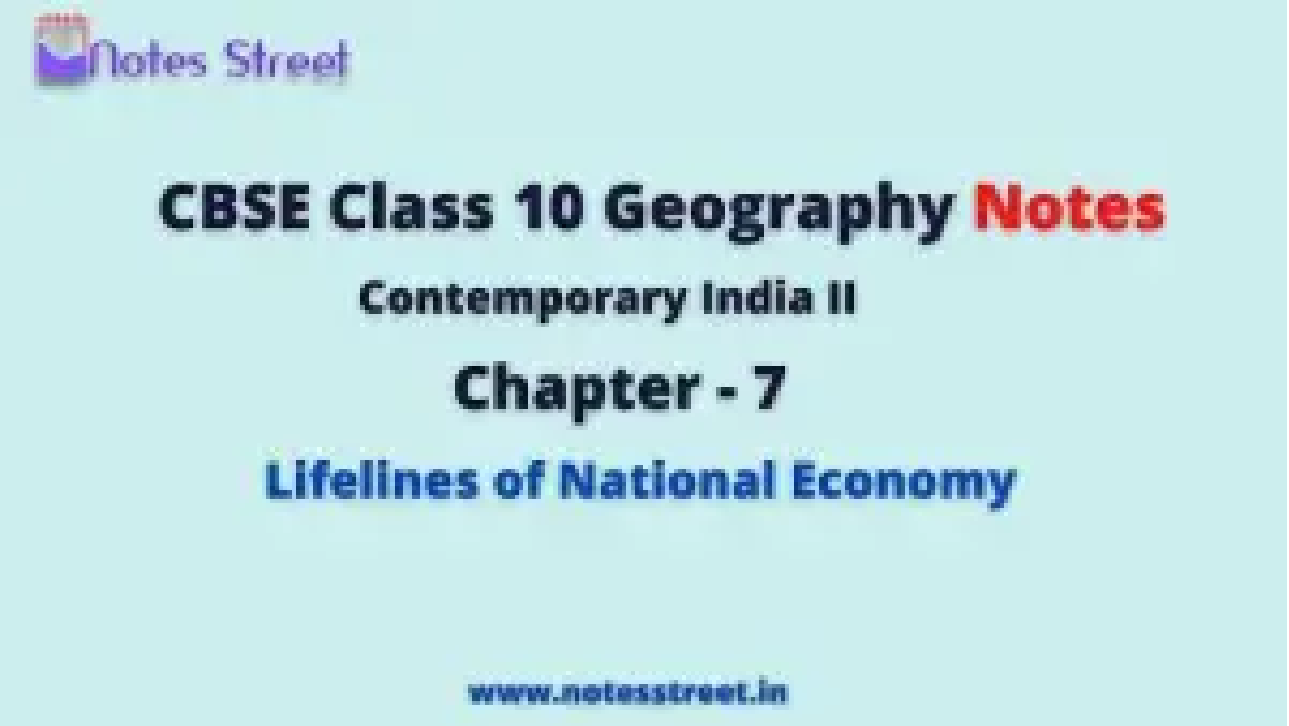

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